

UTM TENSILE REPORT — UTM_Test_20260728_200615

Digital image correlation gauge strain · anchor self-calibration · nominal cross-section

47.4 MPa

UTS

47.0 MPa

Yield σ_y

1.88 GPa

Modulus E

6.0 %

Failure ϵ_f

2527 kJ/m³

Toughness

418 N

Preload anchor

TEST SETTINGS

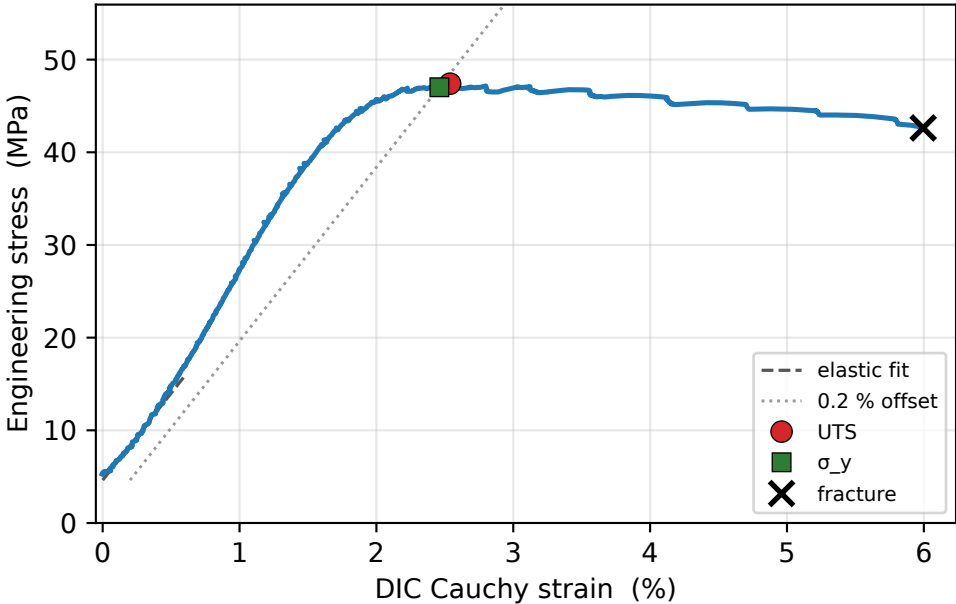
Specimen	UTM_Test_20260728_200615
DIC mode	White
Preload target	470 N
Test speed	0.100 mm/s
Cross-section	80 mm ²
Gauge length L ₀	80 mm
Infill (label)	100 %
Force calib	sc -0.0065 / off 442.318
Comment	—
Test date	2026-07-28 20:04:23
Duration	127.0 s

VALIDATION vs REFERENCES

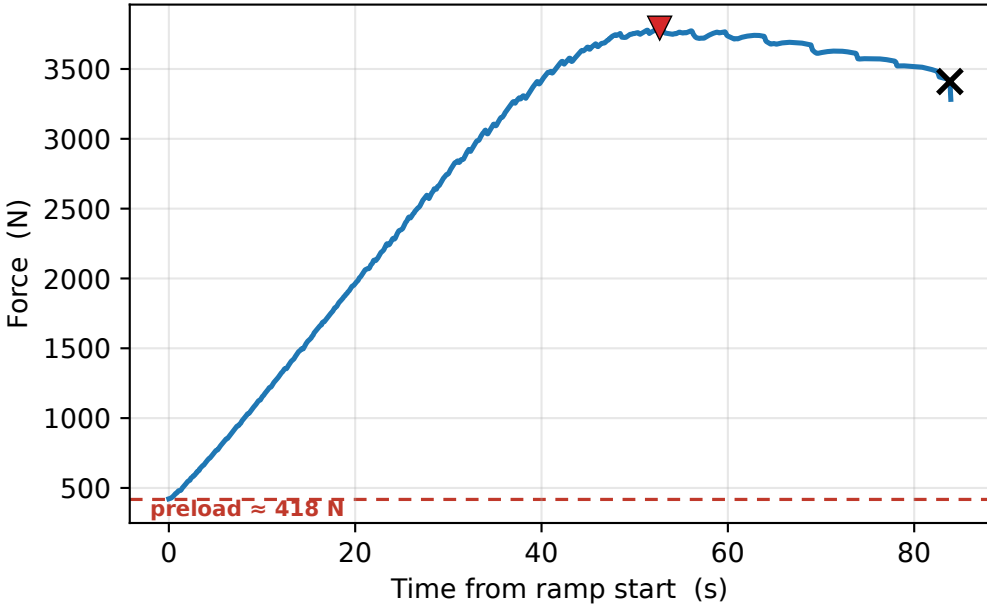
prop	meas	Chacón	E-PLA k
UTS	47.4	32-60 PASS	×1.22
σ_y	47.0	30-50 PASS	×1.23
E	1.9	3-5.5 low	×1.53
ϵ_f	6.0 %	(range 3-7 %)	×1.33

✓ strength inside journal range (k≈1)

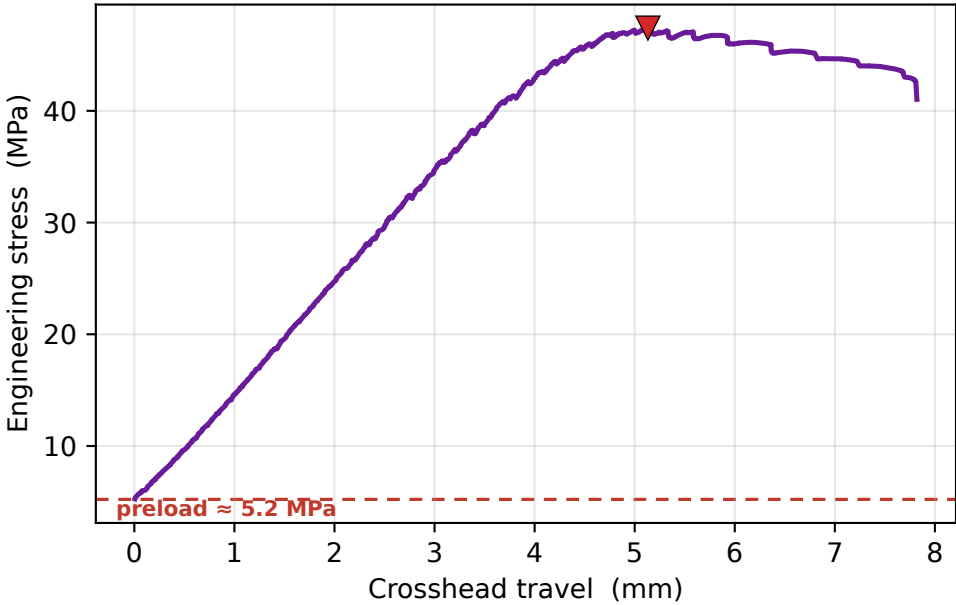
Stress-strain



Load vs time



Stress vs crosshead



Gauge strain vs crosshead

